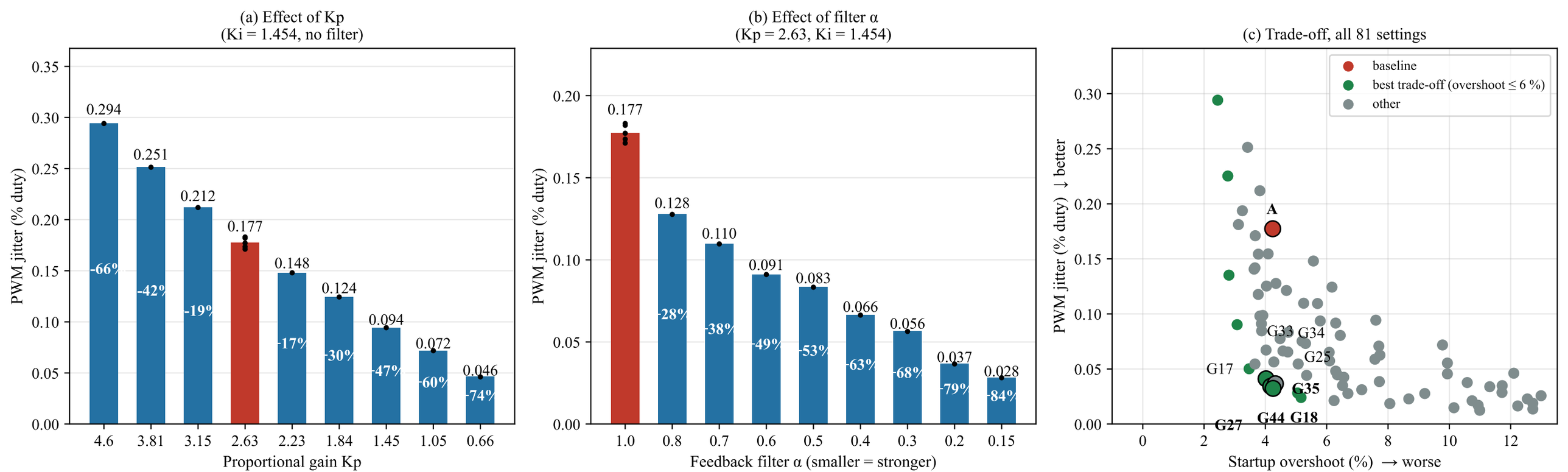
(a) Effect of Kp
(Ki = 1.454, no filter)
PWM jitter (% duty)
0.294
0.251
0.212
0.177
0.148
0.124
0.094
0.072
0.046
-66%
-42%
-19%
-17%
-30%
-47%
-60%
-74%
4.6
3.81
3.15
2.63
2.23
1.84
1.45
1.05
0.66
Proportional gain Kp
(b) Effect of filter α
(Kp = 2.63, Ki = 1.454)
PWM jitter (% duty)
0.177
0.128
0.110
0.091
0.083
0.066
0.056
0.037
0.028
-28%
-38%
-49%
-53%
-63%
-68%
-79%
-84%
1.0
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.15
Feedback filter α (smaller = stronger)
(c) Trade-off, all 81 settings
baseline
best trade-off (overshoot ≤ 6 %)
other
PWM jitter (% duty) ↓ better
A
G33
G34
G25
G17
G35
G27
G44
G18
Startup overshoot (%) → worse